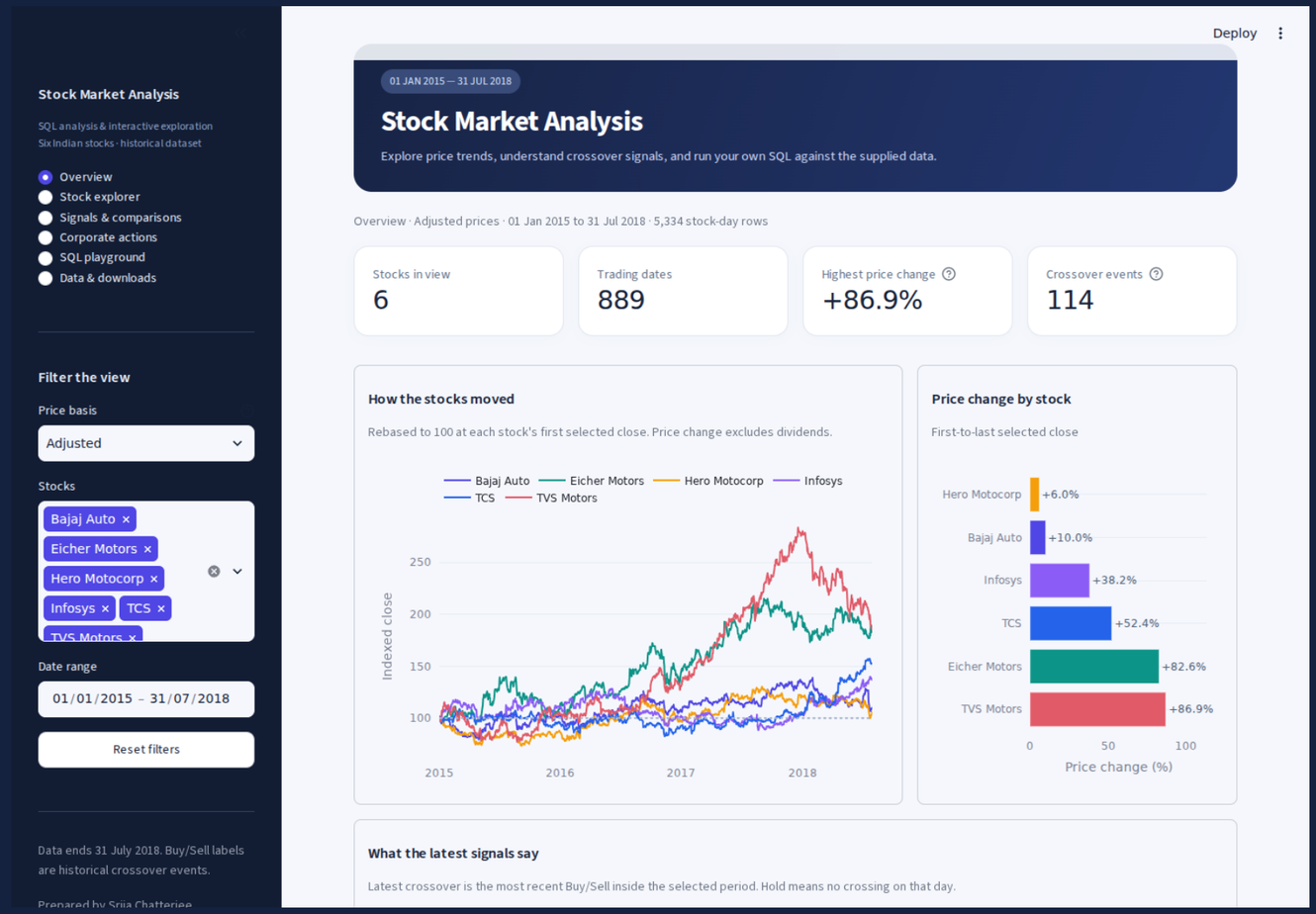
Stock Market Analysis

My SQL analysis and Streamlit dashboard

Srija Chatterjee

1 January 2015 to 31 July 2018
Submission: 10 October 2026



What I set out to do

I wanted to compare the six stocks, explain the signals and check whether the price changes were fair comparisons.

ID	SQL analysis	Check or output
01	History coverage	889 trading dates
02	Five highest Eicher closes	Peak: INR 32,786.40
03	TCS average by year	2018 partial year: INR 2,729.12
04	Missing delivery records	Six rows kept as NULL
05	Bajaj 20/50-day averages	Full windows only
06	Join six closes by date	889 master-table rows
07	Create Bajaj signals	Buy, Sell and Hold

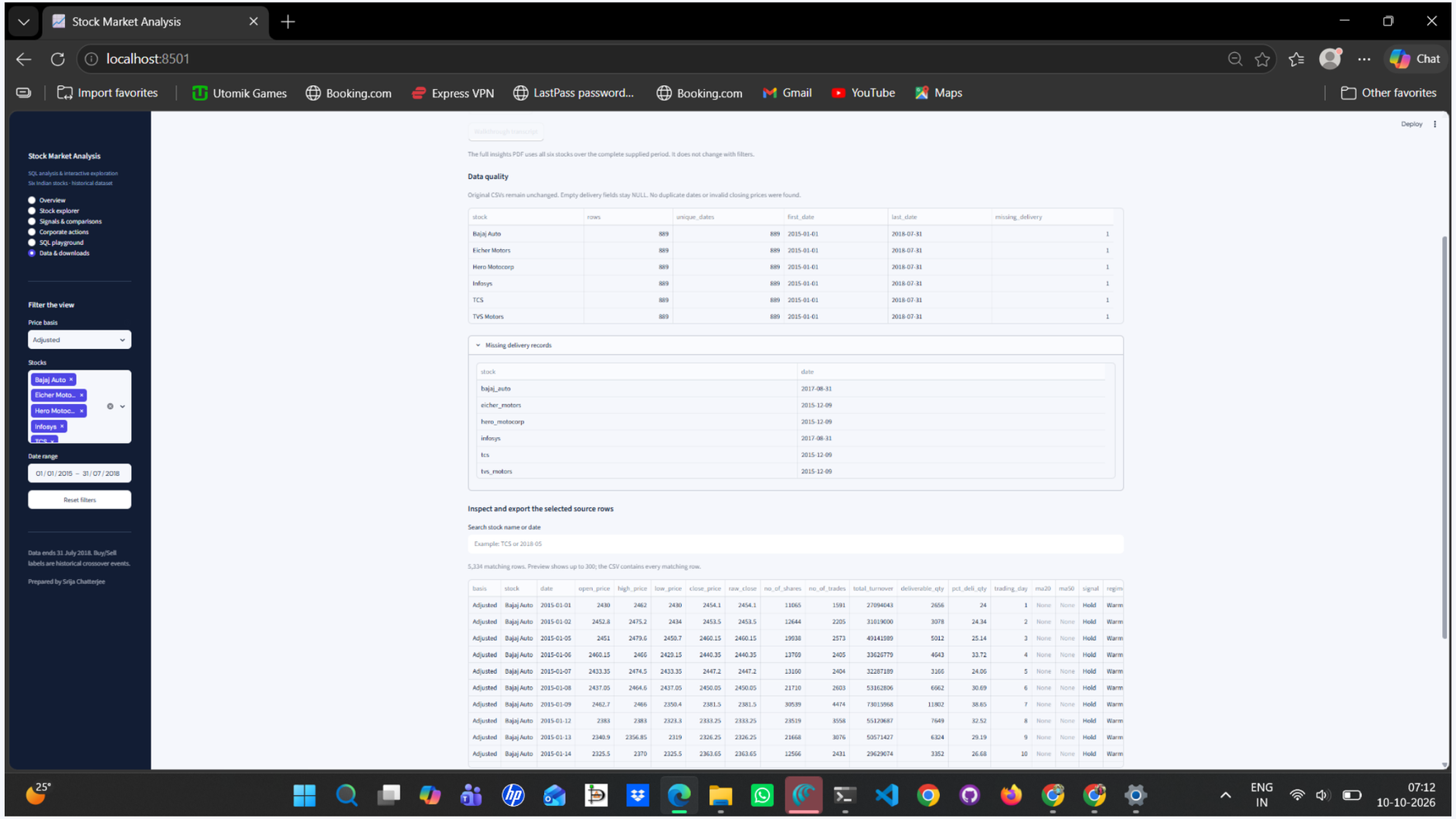
ID	SQL analysis	Check or output
08	Count Bajaj signals	12 Buy, 11 Sell, 866 Hold
09	Bajaj signal function	21 Jun 2018: Buy
10	Six-stock signal summary	Raw: 56 Buy, 57 Sell
11	Full-period raw change	100 × (last / first − 1)
12	Worst daily raw move	IT price cliffs near −50%
13	Bonus-adjusted IT change	Infosys +38.17% TCS +52.36%

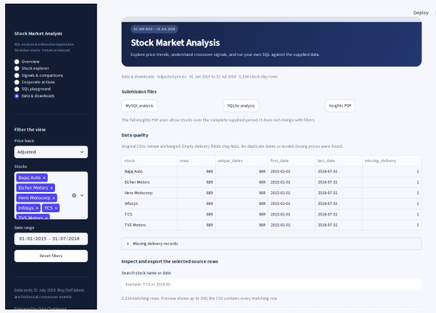
Required submission: MySQL SQL file, insights PDF and a short walkthrough video.

I also added adjusted signals, quick reversals, volatility, drawdown and an execution scenario.

The data I worked with

Six original CSV files contain 5,334 stock-day records over the same 889 trading dates.





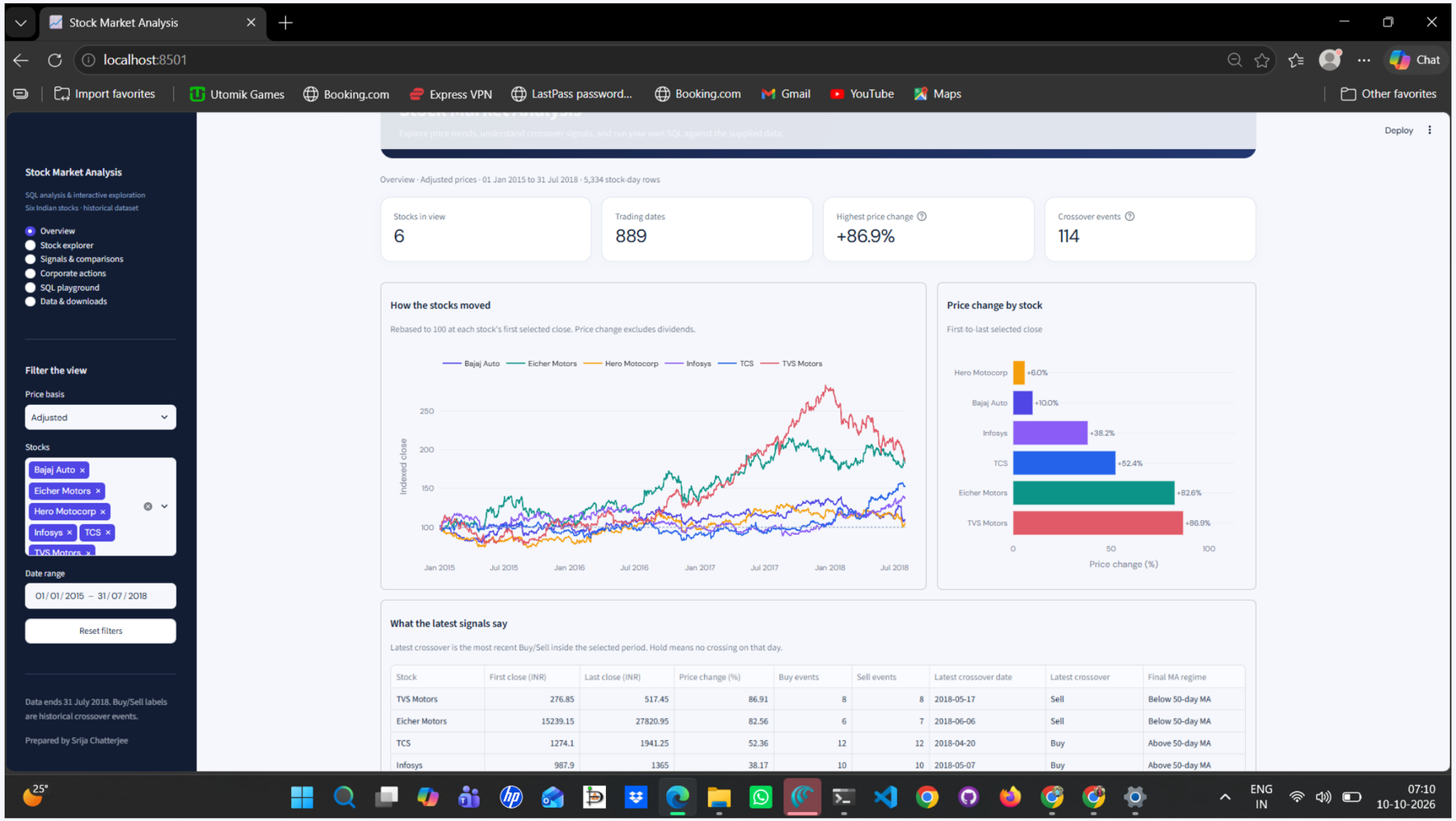
My data checks

- I kept the original files and their missing values.
- Dates are unique and closing prices are complete.
- Each stock has one missing delivery record.
- I can search rows and export the full selection.

Stocks: Bajaj Auto, Eicher Motors, Hero Motocorp, Infosys, TCS and TVS Motors.

The dashboard at a glance

I can compare stocks on the same starting scale and inspect the latest crossover in the selected period.

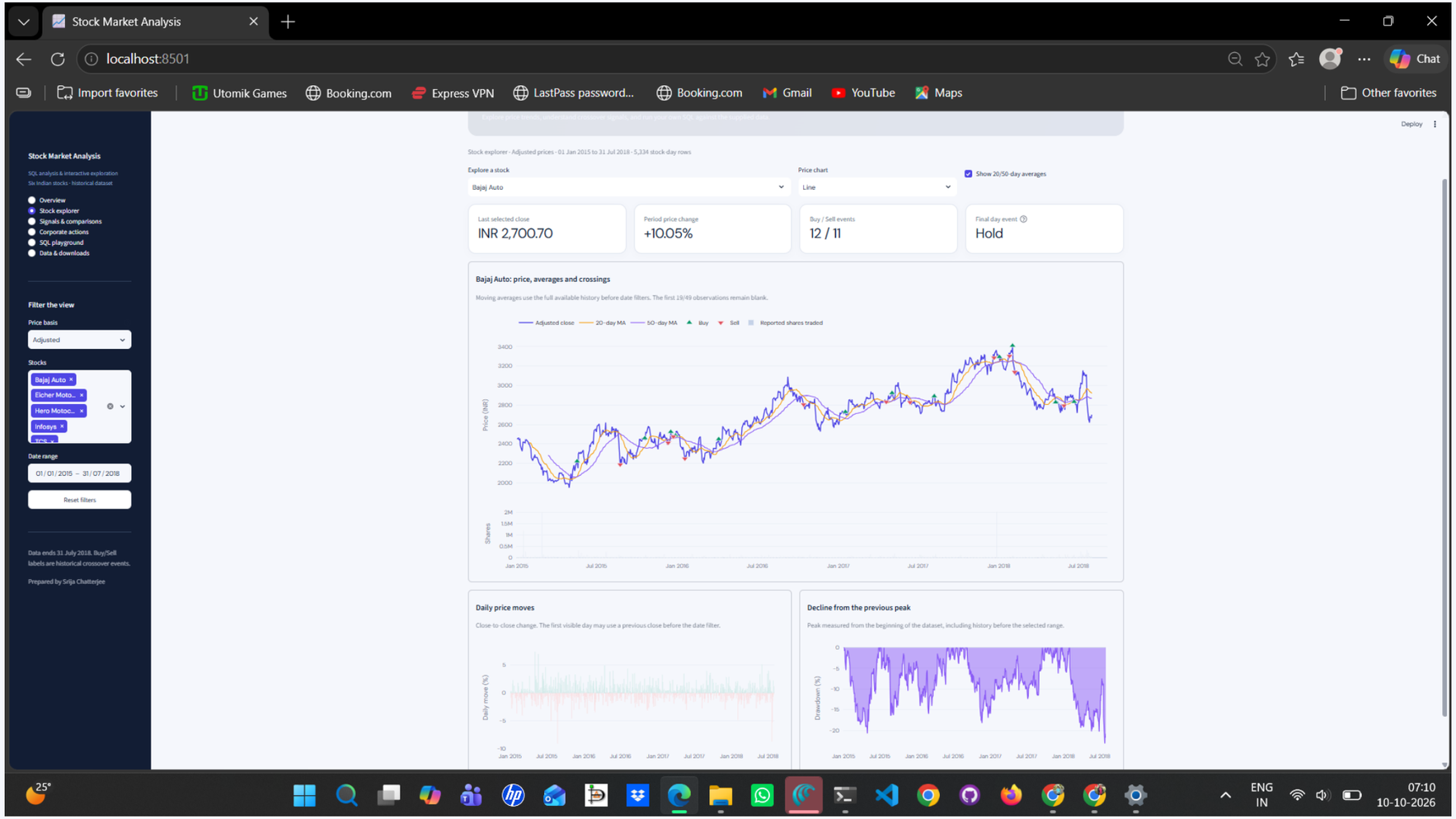


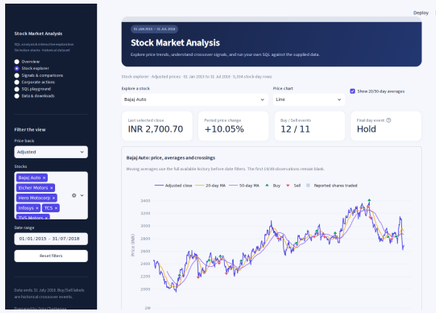
Filters and overview

- Choose stocks, dates and raw or adjusted prices.
- Reset the filters in one click.
- Rebased prices start at 100.
- TVS gained 86.91%, the highest in this period.
- Adjusted data has 114 crossover events.
- The summary includes the latest crossing and final MA regime.

How I explored each stock

The explorer brings the price, moving averages, crossover markers and reported trading volume into one view.





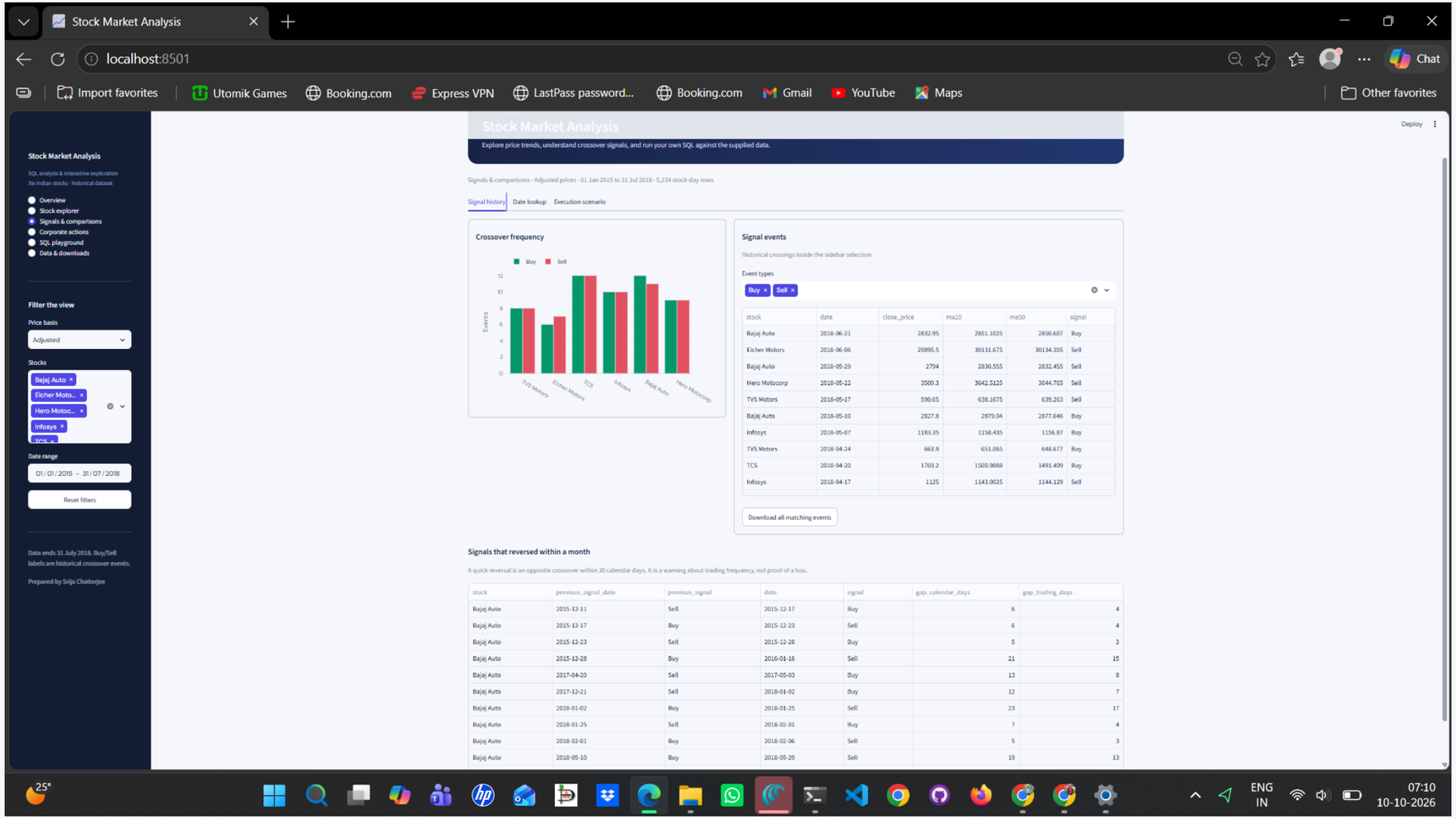
My analysis rules

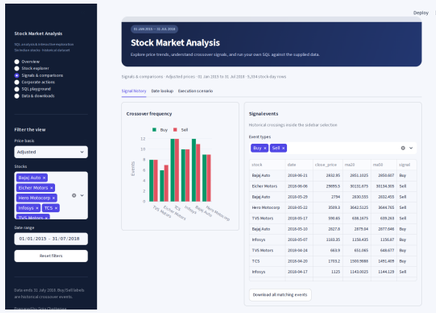
- MA20: today plus 19 earlier trading rows.
- MA50: today plus 49 earlier trading rows.
- The first 19/49 averages stay NULL.
- Line/candlestick switch and MA toggle.
- Daily moves, drawdown and CSV export.

Bajaj checks: first MA20 = INR 2,415.53 on 29 Jan 2015. First MA50 = INR 2,283.80 on 13 Mar 2015.

What Buy, Sell and Hold mean

I compare today’s 20/50-day relationship with yesterday’s relationship to identify a crossing.





The crossover rule

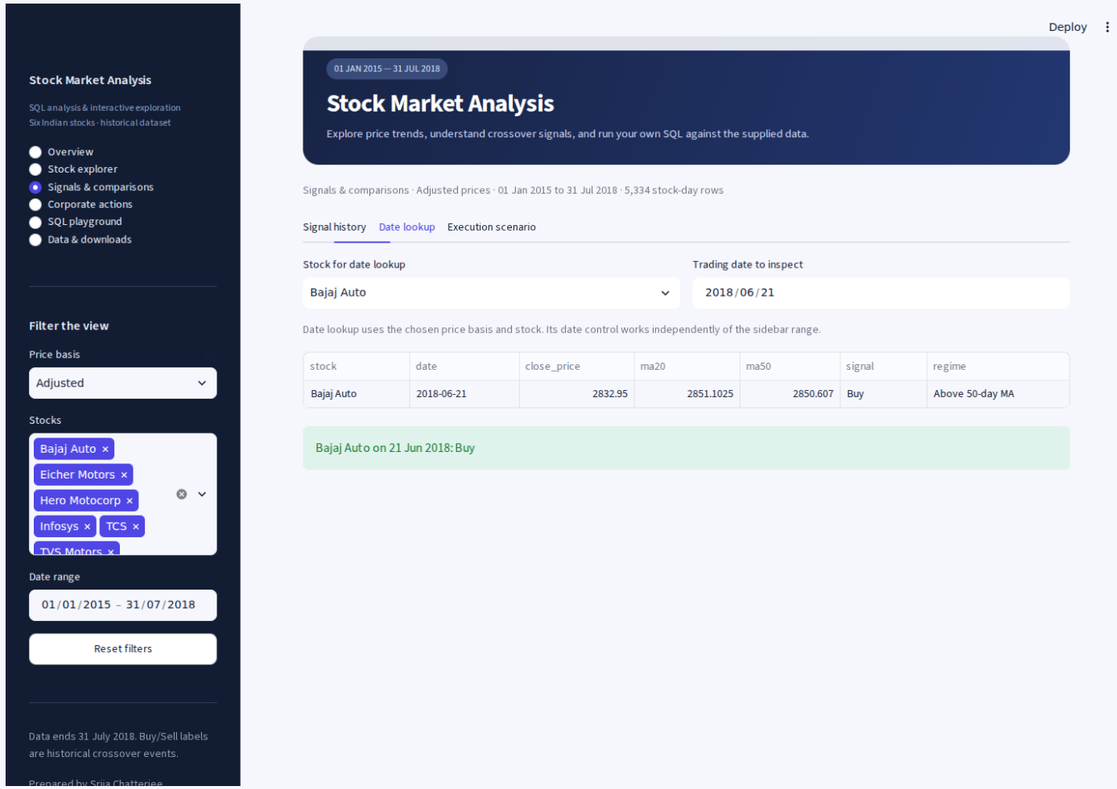
- Buy: MA20 crosses above MA50.
- Sell: MA20 crosses below MA50.
- Hold: no new crossing.
- Raw Bajaj: 12 Buy, 11 Sell.
- Filter events; export CSV.
- Quick reversal: opposite event within 30 days.

A Hold label describes a daily event. The latest crossover and final MA regime describe different parts of the history.

Date lookup and a trading-cost scenario

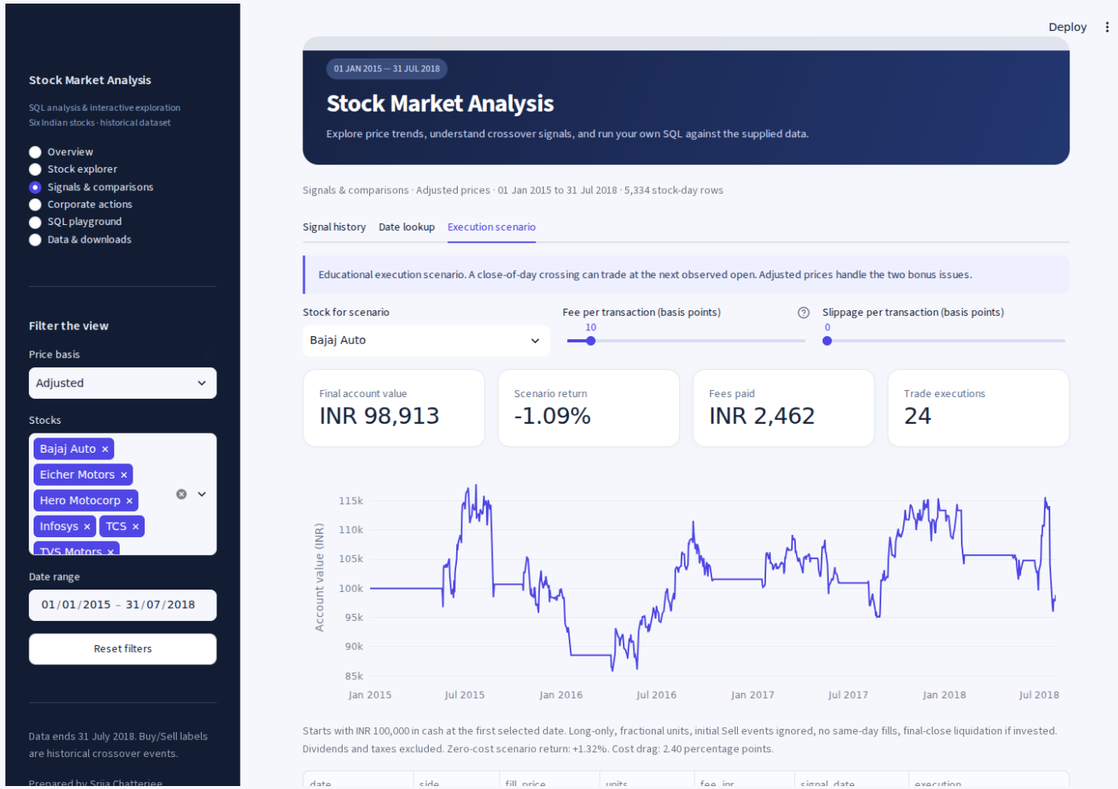
I can inspect a single trading date and test how execution timing and assumed costs affect the result.

Bajaj signal on a chosen date



bajaj_signal('2018-06-21') returns Buy. A missing trading date returns no record or NULL.

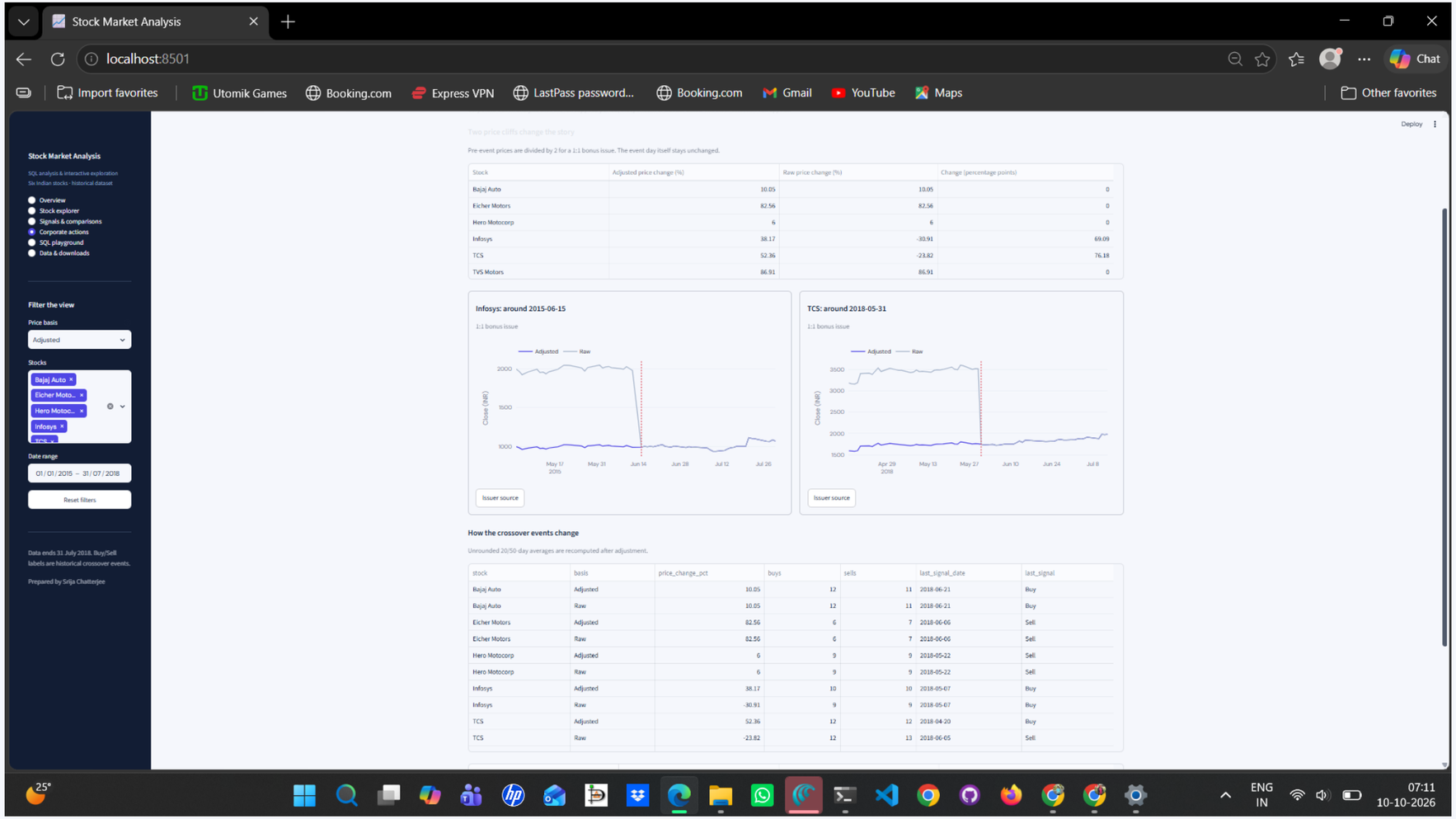
A separate educational scenario



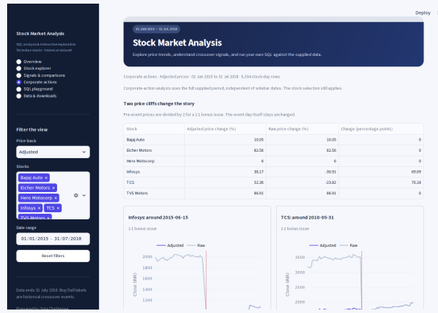
INR 100,000 starting cash. Next observed open. Fee: 0.10% per transaction. Bajaj net return: −1.09%.

Why I adjusted Infosys and TCS

The two 1:1 bonus issues create raw price falls near 50%, which change the return comparison and some signals.



Issuer evidence: Infosys Q1 FY2016, page 5. TCS Annual Report 2018-19, printed page 89.

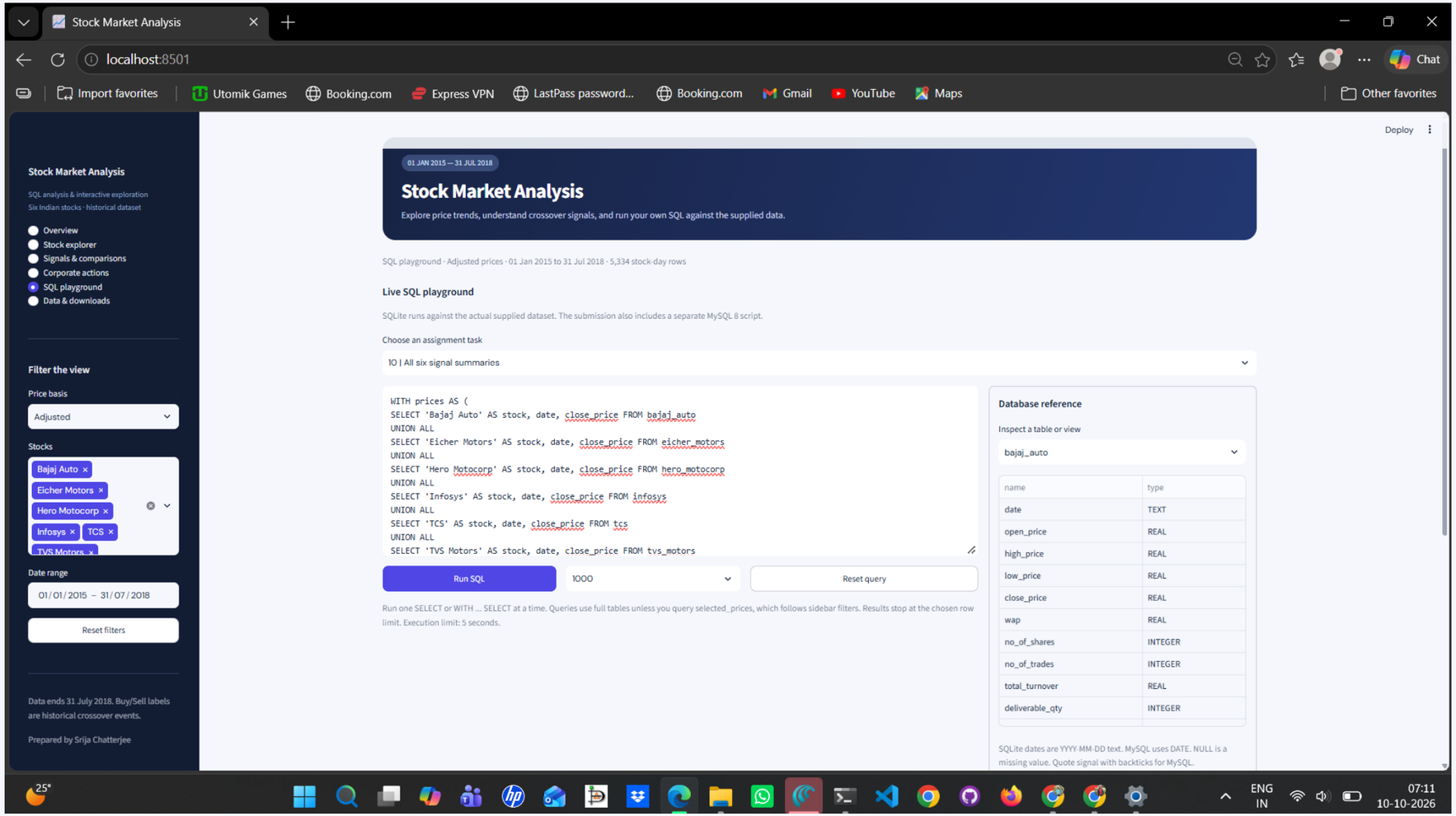


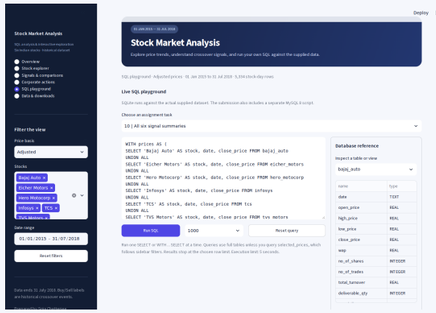
The corrected view

- Infosys: 15 Jun 2015 ex-date.
- TCS: 31 May 2018 ex-date.
- Pre-event OHLC prices ÷ 2.
- Infosys: −30.91% to +38.17%.
- TCS: −23.82% to +52.36%.
- Five signal dates change.

The SQL work behind the dashboard

I can edit a task query, inspect the schema and run it against the actual supplied data.





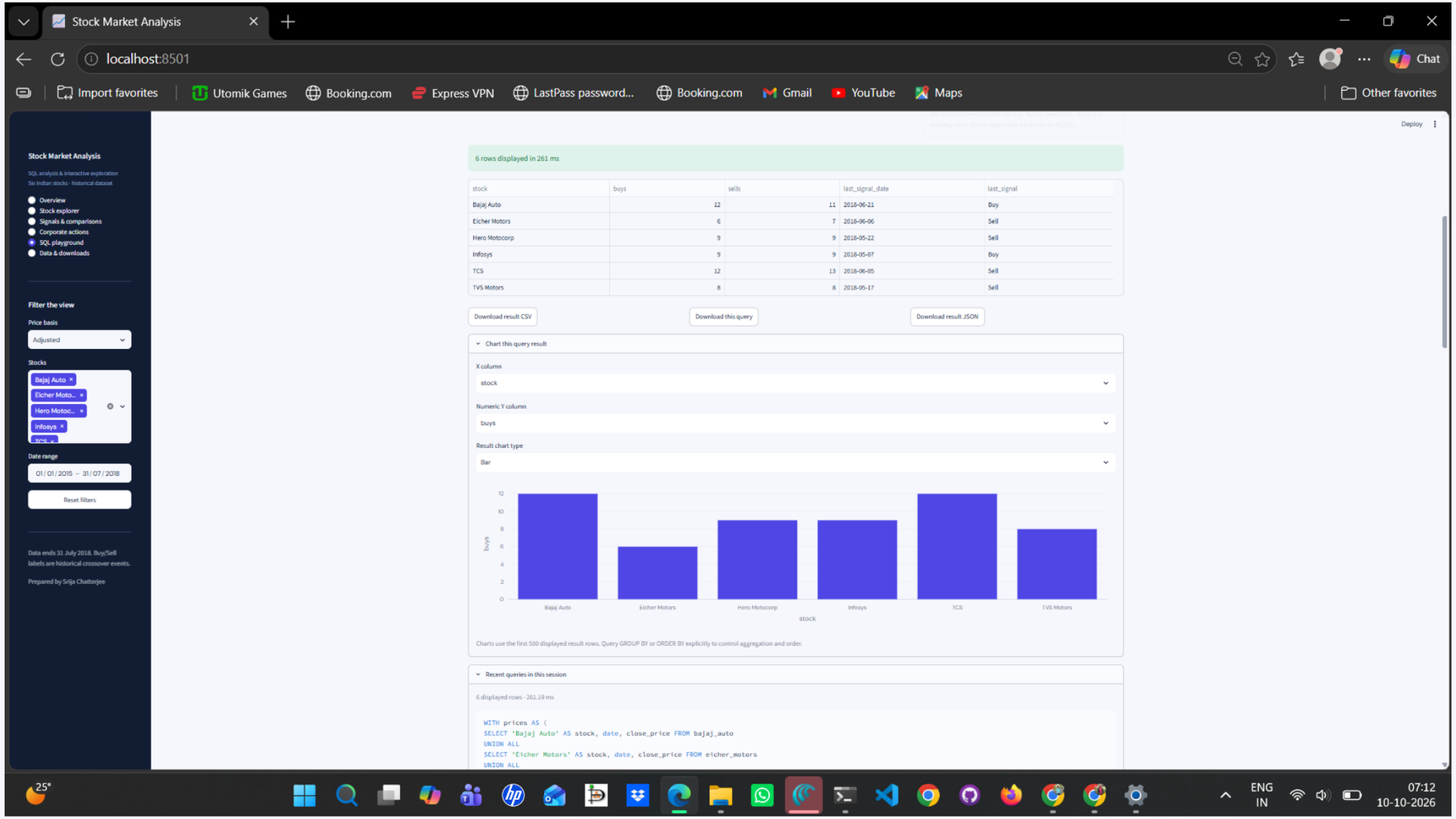
SQL I used

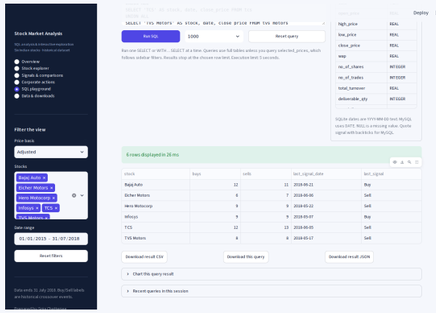
- COUNT, MIN/MAX, AVG: coverage and means.
- ROW_NUMBER, window AVG: complete MAs.
- LAG and CASE: crossing labels.
- JOIN, UNION ALL, CTEs: combine stocks.
- SUM, ROUND, SQRT: counts and risk.

The playground runs one read-only SELECT or WITH … SELECT, with a result row cap and a five-second execution limit.

Query results I can inspect and share

The executed query returns a real result table, and I can make a chart from its numeric columns.





Results and exports

- Raw task 10 returns six rows.
- Total: 56 Buy and 57 Sell.
- CSV, JSON and SQL downloads.
- Choose X, Y and bar/line/scatter.
- Ten recent queries per session.

Task 10 reads the raw assignment tables even when the sidebar says Adjusted. Use selected_prices to follow the sidebar.

What I found across the six stocks

The adjusted results separate full-period price changes from the latest crossing and the cost scenario.

Stock	Price change	Buy / Sell	Latest adjusted crossover	Quick reversals	Net scenario return	Maximum drawdown
Bajaj Auto	+10.05%	12 / 11	Buy 21 Jun 2018	11	-1.09%	-23.18%
Eicher Motors	+82.56%	6 / 7	Sell 06 Jun 2018	1	+71.03%	-28.32%
Hero Motocorp	+6.00%	9 / 9	Sell 22 May 2018	4	-18.68%	-26.90%
Infosys	+38.17%	10 / 10	Buy 07 May 2018	6	+5.27%	-31.06%
TCS	+52.36%	12 / 12	Buy 20 Apr 2018	8	-26.55%	-24.13%
TVS Motors	+86.91%	8 / 8	Sell 17 May 2018	4	+90.89%	-34.59%

TVS and Eicher gained most overall. Their latest Sell crossings indicate later weakness.

TVS also had the deepest adjusted drawdown, −34.59%, and the highest annualized volatility, 30.80%.

Scenario: next observed open, INR 100,000 cash, 0.10% fee each transaction, zero slippage, final-close liquidation. Dividends and taxes excluded.

What I learned and what I would improve

The main lesson for me was to understand the calculation before drawing a conclusion from a chart.

What I learned

- I learned to use SQL windows and previous-day values without using future rows.
- Keeping raw and adjusted prices separate helped me explain the bonus issues.
- I learned to check dates, NULLs and crossing rules before trusting a result.
- The cost scenario showed me why a Buy signal alone cannot establish profitability.
- Connecting each chart to a query made the findings easier to explain.

What I would improve next

- I would add dividends and a broad market benchmark for a fairer comparison.
- I would include a more complete corporate-action history.
- I would test other MA windows on unseen periods and compare realistic costs.
- I would connect fresh data and deploy the dashboard with scheduled updates.
- I would add custom-file validation and more accessibility options.

For this version, my conclusions stay within the six supplied stocks and the 2015-2018 period.